

Reach and Rescue: Hurricane Relief

CSCE 247: Software Engineering

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1. Introduction

The app will be a system that allows communities affected by hurricanes to coordinate relief efforts. It will have specialized tools for medical professionals, volunteers, and victims to give them each access to the resources they need. It will provide up-to-date information on which regions are being hit the hardest and where help is available. It will assist volunteers with resource distribution and shelter management, and victims with finding available shelters and nearby hospitals capable of taking on more patients.

The goal of the app is to assist communities in coping with hurricane damage and help them utilize the resources they need. There aren't many existing platforms with this purpose, and those with similar focuses tend to primarily give broad information about natural disasters. This app would place its focus on specific communities, so that users are shown information that directly pertains to their situation and region.

2. Stakeholders

- First Responders
- Volunteers
- Hurricane Victims

Hannah Wells

Volunteer



Age: 28
Job: 4th Grade Teacher
Address: 468 Glenwood Blvd.

Bio

Hannah Wells is a teacher at Caldwell Elementary School, which has recently been turned into a shelter following the hurricane. The school and her home were on the outskirts of the affected area, so she suffered minimal damage. She has volunteered at the shelter to help people who were misplaced. She wants a system that connects her shelter with the resources it needs and the people who need it.

Skills

- CPR certified
- Leadership
- Can provide emotional support
- Well organized/ can manage the shelter/resources
- High stamina

Resources

- Knowledge of the school
- Personal relationships with the affected

Degree of Distress

☐ 1 Okay	☒ 2 Minimal	☐ 3 Moderate	☐ 4 Troubled	☐ 5 Hazardous
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Necessities / Problems

Easy access to resources

Timeline

Became a volunteer on the third day following the hurricane

Availability

Very open since the school has been shut down

Technology Skill

Average; she grew up with internet access but doesn't have any strong experience

Kyle Donaldson

First Responder



Age: 33
Job: EMT
Address: 2301 High St,
Columbia, SC

Bio

Kyle Donaldson is an EMT working for Prisma Health Hospital in Columbia, SC. Since the hurricane's landfall, Prisma's ER and EMT teams have been working extra time around the clock to aid everyone injured in the storm. To reduce his group's workload, Kyle wants a unified relief system that can be used by all the hospitals across the region to distribute the workload while ensuring all affected victims are treated effectively and efficiently.

Skills

- Mastery of CPR and BLS techniques
- Can make quick decisions
- Adaptable
- Able to accurately document
- Effective at handling stress

Resources

- Knowledge of the hospital system
- Able to respond quickly

Degree of Distress

☐ 1 Okay	☐ 2 Minimal	☒ 3 Moderate	☐ 4 Troubled	☐ 5 Hazardous
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Necessities / Problems

Needs coordination between hospitals

Timeline

began work as an EMT seven years ago w/
no prior hurricane experience

Availability

Low due to large number of emergency
calls

Technology Skill

Above average; understands medical
technology

Katrina Kane

Citizen in need



Age: 47

Job: Local restaurant owner

Address: 1234 Generic Ave

Bio

Katrina has owned and operated a local restaurant with her family for over a decade. When the hurricane hit, her building had a few trees fall on it, breaking walls, roofing, and windows. The bottom floor, containing her restaurant, is flooded. Katrina and her family live on the top floors. Right now, her place is unlivable. She and her family have been staying at a shelter nearby, but her mom is ill and needs constant treatment and care.

Skills

- Cooking in large amounts
- organizing and cleanliness
- personable
- comfortable delegating tasks
- inventory count

Resources

- restaurant staff

Degree of Distress

☐ 1 Okay	☐ 2 Minimal	☐ 3 Moderate	☐ 4 Troubled	☒ 5 Hazardous
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Necessities / Problems

Can't find other locations for her and her family to stay at safely, and to give her mom the medical help she needs.

Timeline

She and her family were misplaced during the storm

Availability

Moderate. She is not currently working, but she has to take care of her mom and kids.

Technology Skill

Below average, comfortable with basic applications, but not tech-savvy beyond that

3. Constraints

Time Constraints

- Project must be completed by Dec 5th, 2026

Monetary Constraints

- The project must be developed at the cost of \$0

Technical Constraints

- The code must be written in Java
- The website / app must be accessible on multiple types of devices (laptop, PC, mobile devices)

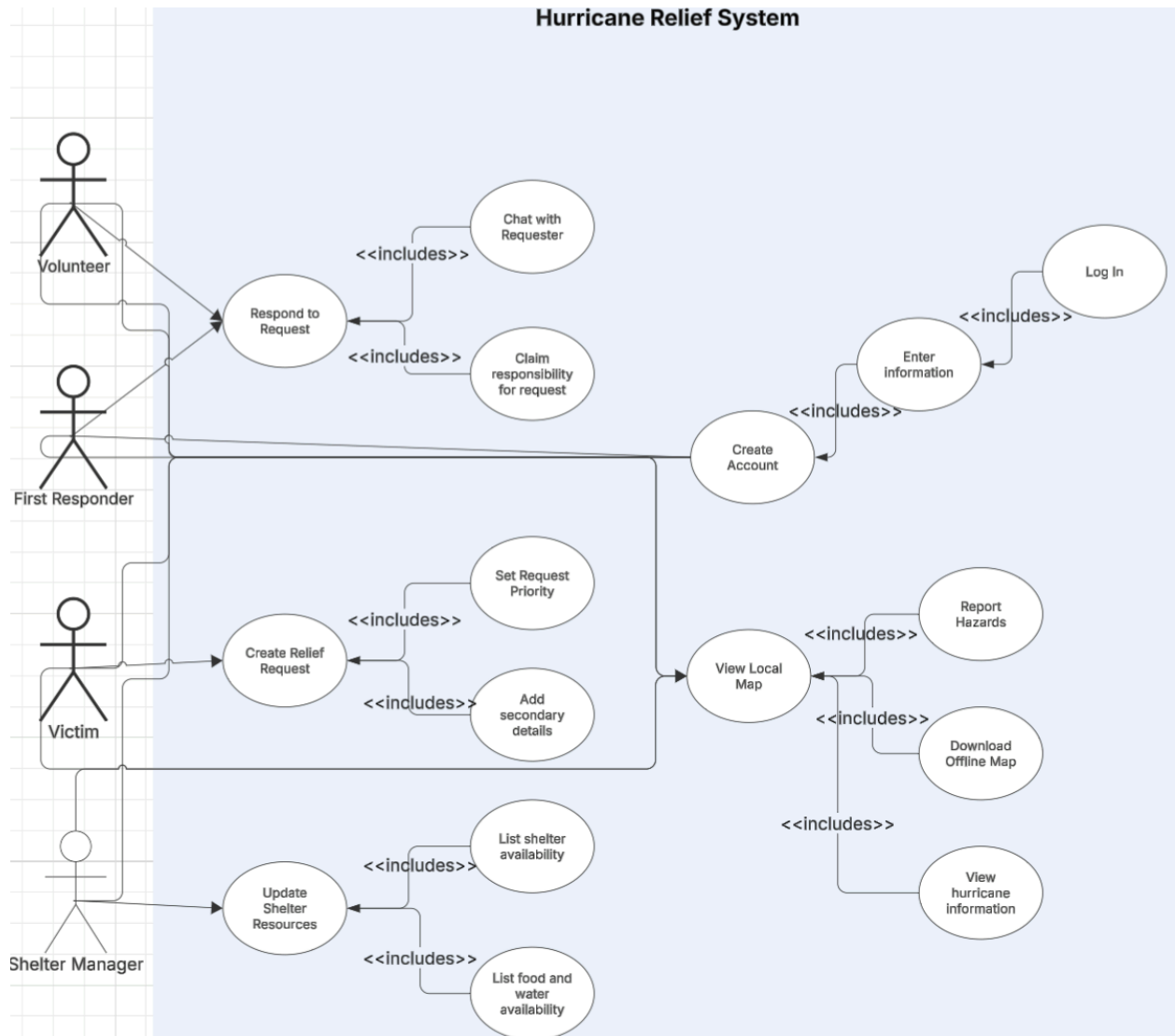
Accessibility Constraints

- Users must be able to easily access the features on both the web and the app
- Software must adapt to laptops, PC, and mobile devices
- Must be easy to navigate for people with little technical experience

4. Overall Description

This app must be accessible to users on both mobile devices and PCs/Laptops with or without a WiFi connection. The app will collect real-time data about oncoming, active, and recent hurricanes in a nearby geographic area relative to the user. The app must allow responders to act with a higher level of permissions to monitor requested aid, available volunteers, levels of resources to be distributed, and capacity of shelters. Users must be able to use the app, either with an account or anonymously, to request aid, list their level of distress, list available skills/resources, and place themselves as a victim or volunteer. The app will contain a database of storm information, shelters, resources, volunteers, places of distress, user location, and timeline of events. The app will have a public message board to communicate with other users and private messaging to directly contact each other.

5. Business Use Cases



6. Functional Requirements

[Requirements Spreadsheet](#)

7. Non-Functional Requirements

Look and Feel Requirements

- The user interface of the system shall be implemented in a way for all types of users, such as beginners, elderly, or disabled/injured users, to understand and utilize the functionality of the app
- Icons should be big, simple, and brightly visualized

Usability Requirements

- The system shall be used by users with a wide range of capabilities
- Text to Speech, Screen Reader, Voice Control, and Alternate Image Text will all be implemented to help anyone no matter the circumstance

Performance Requirements

- The system shall be able to run on any device that is connected to the internet
- Certain select features will be accessible without internet or cellular connection

Maintainability and Support Requirements

- The system shall be able to run on Window, Linux, and Mac systems

Security Requirements

- The user's information shall be stored in a secure manner
- Background checks will be performed on volunteers to ensure hurricane-victim safety
- All medical credentials will be checked to ensure validity

Legal Requirements

- The system shall comply with all state and local law

8. Definitions and Acronyms

- POI: Point of Interest

9. Competitive Analysis

Hurricane Preparedness by American Red Cross		
Strengths	Weaknesses	Focus
- Real-time weather service alerts are customizable by location and hazard type - Live hurricane tracking and interactive maps - Step-by-step preparedness guidelines and checklists - "I'm safe" messages let users notify loved ones they are ok even if the other party does not have the app - Preloaded content allows users to work offline for guidance content - Built-in screen reader - Overrides do not disturb for alerts	- complaints about missed alerts notifications not sending to device - prep/response focused, not recovery/aid - doesn't manage needs or aid registration for volunteer matching	This app prepares individuals in any way they can before a hurricane. It alerts the user of hazards and tells them how to be safe.

Local Relief		
Strengths	Weaknesses	Focus
- Relatively accessible - Non-profit - Anyone can post resources they find - Covers a wide variety of regions and types of disasters - Allows users to share what has helped them deal with a crisis to help their community.	- Small user base - Somewhat difficult to navigate - Unpredictable update schedule - Relies on a few moderators to post the bulk of the updates. - Reports of the system being buggy	Sharing resources with people in crisis areas. It is formatted like a social media platform, but for sharing updates/resources for disaster relief.

FEMA		
Strengths	Weaknesses	Focus
- Real-time weather alerts - Shelter and disaster recovery center locator and personalized map showing the closest facilities (with text message shelter lookup backup) - Lets user upload photos of damage for recovery efforts - Recovery FAQ - How to build a family communication plan and emergency kit	- No live hurricane tracker - Cannot submit disaster assistance application in the app (only tracks status) - Has needed several rereleases to fix core navigation issues	FEMA focuses on the before, during, and after of a hurricane. The app helps guide users in preparedness and provides shelter alerts. And checks eligibility for assistance after the hurricane while allowing users to report damage.

Summary	Strengths	Weaknesses	Focus
Hurricane Preparedness by American Red Cross	+ realtime alerts + live hurricane tracking +preparedness guidelines -offline access	- no recovery center locator -no disaster assistance	-preparedness
Local Relief	+ Lets users share what has helped during the crisis +recovery resources +organization provided support	- difficult to navigate -no weather tracking or alert feature	-community driven peer to peer coordination (local social layer)
FEMA	+ Real-time alerts + personalized map + sms fallback +disaster tracking/reporting +preparedness checklist	- no live hurricane tracker -no requester, volunteer matching -no offline access	-guides users on what to do after the hurricane

Complete Summary:

Across all of the apps, disaster response splits into 3 main pieces: official real-time awareness, recovery/assistance, and community coordination. There is a clear gap among them

because no single app does all three. Local relief is the most similar to our idea, showing both that the demand for two-sided matching is real and that it is risky to build without institutional-grade reliability. We want to let communities affected by hurricanes coordinate relief.

While the Red Cross and FEMA broadcast information outward and Local Relief lets people post loosely, our design will have tiers. We are hoping to combine the organization and clear structure of the former two with the user control and community sourcing that Local Relief allows. Responders and general users are allowed to make structured aid requests, and are given skill/resource listings and a live database of shelters, resources, and distress levels. This is built specifically for the gap that none of the competitors close, using structured, region-specific data and role permissions instead of a general information feed or an unmoderated social board.

10. References

List any documents referred to in the creation of this requirements document

11. Appendices

Any relevant information that can assist in understanding the requirements